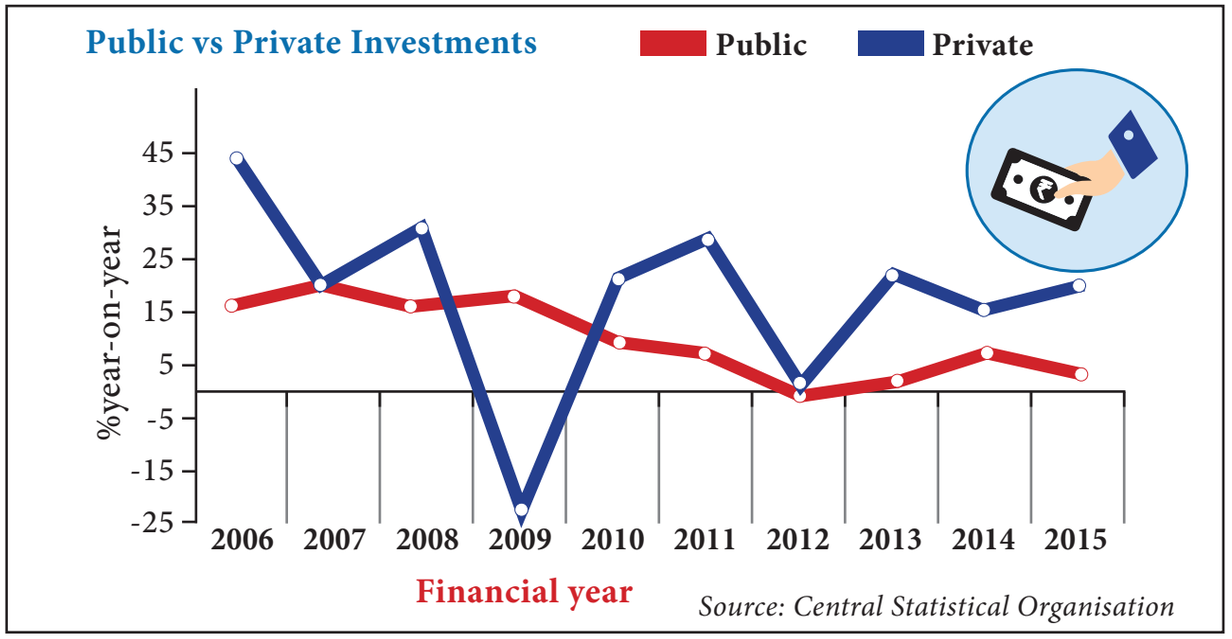


Source: Central Statistical Organisation

3. Karl Pearson's Coefficient of Correlation

Karl Pearson's Method is popularly known as Pearson's coefficient of correlation denoted by the symbol 'r'. The coefficient of correlation 'r' measures the degree of linear relationship between two variables say X and Y. The Formula for computing Karl Pearson's Coefficient of correlation is:

1) $$r = \frac{N\Sigma XY - (\Sigma X)(\Sigma Y)}{\sqrt{N\Sigma X^2 - (\Sigma X)^2}\sqrt{N\Sigma Y^2 - (\Sigma Y)^2}}$$

'r' is calculated by **Direct Method** without taking deviation of terms either from actual mean or assumed mean.

2) r is calculated by taking the **Deviation from actual mean.**

$$r = \frac{\Sigma xy}{\sqrt{\Sigma x^2 \Sigma y^2}}$$ where $x = (x-\bar{x})$, $y = (y-\bar{y})$

3) 'r' is calculated by taking assumed mean

$$r = \frac{N\Sigma dxdy - (\Sigma dx)(\Sigma dy)}{\sqrt{N\Sigma dx^2 - (\Sigma dx)^2}\sqrt{N\Sigma dy^2 - (\Sigma dy)^2}}$$

Where dx refers to deviations of x series from assumed mean A (X-A), dy refers to deviations of y series from an assumed mean θ (Y-B)

$\Sigma dxdy$ = Sum of product of the deviations x and y series from their assumed means.

Σdx^2 = Sum of the squares of the deviations of x series from an assumed mean

Σdy^2 = Sum of the squares of the deviations of y series from an assumed mean

Σdx = sum of the deviation of x series from an assumed mean of x

Σdy = sum of the deviation of y series from an assumed mean of y

Procedure for Computing the Correlation Coefficient: (For Direct and Deviation from actual mean method).

- **Step-1** Calculate the mean of two series 'X' 'Y'
- **Step-2** Calculate the deviations 'X' and Y in two series from their respective mean.

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